

SYNOPSIS CHAMPIONSHIP PROJECT ABSTRACT

The Abstract is a required part of your project. Bring 1 copy at Project Check-in.

Your abstract should be written after you finish the research and experimentation and should include:

- **Project number**, title and the full name of every team member
- Purpose of the project
- Procedure
- Observations/data/results
- Conclusions
- 250 words or less (to fit on one page)
- Use a black font size 10 to 12

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(Project Number: P99 (

Project Title: Online Learning of Smooth Functions

Online learning is a framework in which a machine learning system makes a prediction, observes the correct outcome, and immediately updates its internal model, processing data sequentially rather than in a single batch. This approach is widely used in real-time and resource-constrained settings, where predictions must be made continuously and adaptation must occur without storing or retraining on datasets.

This project investigates how explicit computational limits affect online learning performance. It studies two constraints: an operation cap, which limits the number of operations allowed per prediction, and a memory cap, which limits the amount of stored information. The central question is how much per-round computation is necessary and sufficient bounds for worst-case prediction error when learning smooth functions under adaptive or adversarial input sequences. It establishes upper and lower bounds that quantify how prediction error decreases as the per-round computational budget increases. In particular, for broad classes of smooth functions, it is shown that worst-case uniform error decreases according to a precise power law in the number of allowed operations, and that no algorithm with the same cap can achieve a substantially better rate.

More generally, this project proves an information – computation tradeoff: if a function class is too complex relative to the operation budget, persistent error is unavoidable, whereas sufficiently simple classes can be learned reliably within the same constraints. This also identifies sharp computational thresholds for structured families, including linear transformations and polynomials.

These results provide rigorous performance guarantees for predictive systems operating under computational and memory limitations.